Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

[CGPA-3.32] in Cycle 1

IT13: ADBMS

Chapter 2:SQL

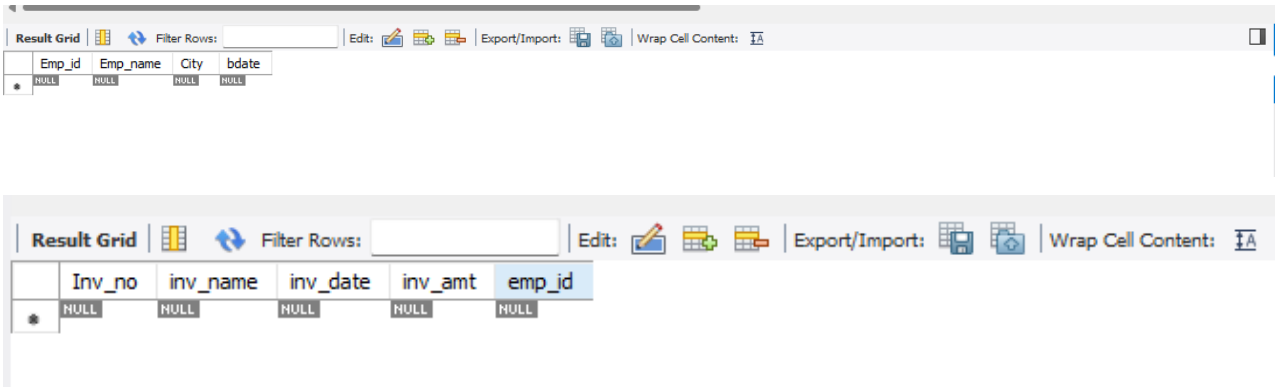
Practical Assignment 1 Solution Template

Write the SQL query and paste a screenshot of the output below the question.

1. Create a database and two tables in it.

   Solution:

```
1  create database company_db;
2  show databases;
3  use company_db;
4
5  create table Employee(Emp_id int primary key,Emp_name varchar(20),City varchar(20),bdate date);
6  create table Investor(Inv_no int primary key, inv_name varchar(20),inv_date date,inv_amt int,emp_id int);
7
```



2. Insert at least 5 rows into the table.

   Solution:

```
10 •  select*from Investor;
11
12 •  insert into Employee values(101,'Sam','Nagpur','2005-05-30');
13 •  insert into Employee values(102,'Khush','Dubai','2020-09-11');
14 •  insert into Employee values(103,'Koko','Chandrapur','2006-06-26');
15 •  insert into Employee values(104,'Adarsh','Nagpur','2011-07-17');
16 •  insert into Employee values(105,'Shreya','Indore','2004-02-16');
17 •  select*from Employee;
18
19 •  insert into Investor values(201,'Sam','2010-03-12',30000,101);
20 •  insert into Investor values(202,'Khush','2015-07-07',50000,102);
21 •  insert into Investor values(203,'Koko','2020-04-15',70000,103);
22 •  insert into Investor values(204,'Adarsh','2025-09-25',80000,104);
23 •  insert into Investor values(205,'Shreya','2026-06-22',40000,105);
24
25 •  select*from Investor;
26
```

Result Grid | Filter Rows: | Edit: | Export/

Emp_id	Emp_name	City	bdate
101	Sam	Nagpur	2005-05-30
102	Khush	Dubai	2020-09-11
103	Koko	Chandrapur	2006-06-26
104	Adarsh	Nagpur	2011-07-17
105	Shreya	Indore	2004-02-16

Employee 25

Result Grid | Filter Rows: | Edit:

Inv_no	inv_name	inv_date	inv_amt	emp_id
201	Sam	2010-03-12	30000	101
202	Khush	2015-07-07	50000	102
203	Koko	2020-04-15	70000	103
204	Adarsh	2025-09-25	80000	104
205	Shreya	2026-06-22	40000	105

Investor 26

3. Display all the information in the Employee table.

   Solution:

Result Grid | Filter Rows: | Edit: | Export/

Emp_id	Emp_name	City	bdate
101	Sam	Nagpur	2005-05-30
102	Khush	Dubai	2020-09-11
103	Koko	Chandrapur	2006-06-26
104	Adarsh	Nagpur	2011-07-17
105	Shreya	Indore	2004-02-16

2020-09-11

Employee 25

Result Grid | Filter Rows: | Edit:

Inv_no	inv_name	inv_date	inv_amt	emp_id
201	Sam	2010-03-12	30000	101
202	Khush	2015-07-07	50000	102
203	Koko	2020-04-15	70000	103
204	Adarsh	2025-09-25	80000	104
205	Shreya	2026-06-22	40000	105

Investor 26

4. Add columns 'Age' and 'Salary' to the Employee table; apply check constraints for (Age<=50 and Salary>=10000).

Solution:

```
26
27 •  alter table Employee add column Age int, add check(Age<=50);
28 •  alter table Employee add column Salary int, add check(Salary>=10000);
29 •  desc Employee;
30
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Table	Create Table
Employee	CREATE TABLE `employee` (`Emp_id` int NO...

```
CREATE TABLE `employee` (
  `Emp_id` int NOT NULL,
  `Emp_name` varchar(20) DEFAULT NULL,
  `City` varchar(20) DEFAULT NULL,
  `bdate` date DEFAULT NULL,
  `Age` int DEFAULT NULL,
  `Salary` int DEFAULT NULL,
  PRIMARY KEY (`Emp_id`),
  CONSTRAINT `emplo...
```

Result 28

Output

Action Output

5. Add a column 'Age' to the Investor table.

Solution:

```
32 •  alter table Investor add column Age int;
33 •  select*from Investor;
```

Result Grid | Filter Rows: | Edit:

Inv_no	inv_name	inv_date	inv_amt	emp_id	Age
201	Sam	2010-03-12	30000	101	NULL
202	Khush	2015-07-07	50000	102	NULL
203	Koko	2020-04-15	70000	103	NULL
204	Adarsh	2025-09-25	80000	104	NULL
205	Shreya	2026-06-22	40000	105	NULL

6. Change the data type of inv_name from varchar to text in the Investor table

Solution:

```
37 •  alter table Investor Modify inv_name text;
```

Result Grid | Filter Rows: | Edit:

Inv_no	inv_name	inv_date	inv_amt	emp_id	Age
201	Sam	2010-03-12	30000	101	NULL
202	Khush	2015-07-07	50000	102	NULL
203	Koko	2020-04-15	70000	103	NULL
204	Adarsh	2025-09-25	80000	104	NULL
205	Shreya	2026-06-22	40000	105	NULL

Investor 31

7. Change the name of the column/field Emp_id to E_id.

Solution:

```
40 •  alter table Employee rename column Emp_id to E_id ;
41 •  select*from Employee;
```

Result Grid | Filter Rows: | Edit:

E_id	Emp_name	City	bdate	Age	Salary
101	Sam	Nagpur	2005-05-30	NULL	NULL
102	Khush	Dubai	2020-09-11	NULL	NULL
103	Koko	Chandrapur	2006-06-26	NULL	NULL
104	Adarsh	Nagpur	2011-07-17	NULL	NULL
105	Shreya	Indore	2004-02-16	NULL	NULL

Employee 32

8. Update the city of Emp_id-4 to the current city, Nagpur.

Solution:

```
update Employee set City='Nagpur' where E_id=103;
select*from Employee;
```

Result Grid | Filter Rows: | Edit:

E_id	Emp_name	City	bdate	Age	Salary
101	Sam	Nagpur	2005-05-30	NULL	NULL
102	Khush	Dubai	2020-09-11	NULL	NULL
103	Koko	Nagpur	2006-06-26	NULL	NULL
104	Adarsh	Nagpur	2011-07-17	NULL	NULL
105	Shreya	Indore	2004-02-16	NULL	NULL

9. Delete one record from the Investor table.

Solution:

```
delete from Investor where inv_no=201;
select*from Investor;
```

Result Grid | Filter Rows: | Edit:

Inv_no	inv_name	inv_date	inv_amt	emp_id	Age
202	Khush	2015-07-07	50000	102	NULL
203	Koko	2020-04-15	70000	103	NULL
204	Adarsh	2025-09-25	80000	104	NULL
205	Shreya	2026-06-22	40000	105	NULL
NULL	NULL	NULL	NULL	NULL	NULL

10. Drop the column 'Age' from the Investor table.

Solution:

```
alter table Investor drop column Age;
select*from Investor;
```

Result Grid | Filter Rows: | Edit:

Inv_no	inv_name	inv_date	inv_amt	emp_id
202	Khush	2015-07-07	50000	102
203	Koko	2020-04-15	7000	3
204	Adarsh	2025-09-25	80000	104
205	Shreya	2026-06-22	40000	105
NULL	NULL	NULL	NULL	NULL

70000

11. Drop the CHECK constraint on the 'Age' column from the Employee table.

Solution:

```
55 •  alter table Employee drop constraint employee_chk_1;
56 •  show create table Employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Table	Create Table
Employee	CREATE TABLE `employee` (`E_id` int NOT N...

```
CREATE TABLE `employee` (
  `E_id` int NOT NULL,
  `Emp_name` varchar(20) DEFAULT NULL,
  `City` varchar(20) DEFAULT NULL,
  `bdate` date DEFAULT NULL,
  `Age` int DEFAULT NULL,
  `Salary` int DEFAULT NULL,
  PRIMARY KEY (`E_id`),
  CONSTRAINT `employee_...
```

Result 36 ×

Output

Action Output

12. Create foreign key constraints between the Employee and Investor tables

Solution:

```
58 •  alter table Investor rename column Emp_id to E_id;
59 •  alter table Investor add foreign key(E_id) references Employee(E_id);
60 •  desc Investor;
61 •  select* from Investor;
62 •  select*from Employee;
63 •  desc Employee;
```

Result Grid | Filter Rows: | Edit:

Inv_no	inv_name	inv_date	inv_amt	E_id
202	Khush	2015-07-07	50000	102
203	Koko	2020-04-15	70000	103
204	Adarsh	2025-09-25	80000	104
205	Shreya	2026-06-22	40000	105
NULL	NULL	NULL	NULL	NULL

Investor 37 ×

13. Display any 4 columns from the Investor table.

Solution:

```
72 •  select inv_no,inv_name,inv_date,inv_amt from Investor;
```

Result Grid | Filter Rows: | Export:

inv_no	inv_name	inv_date	inv_amt
202	Khush	2015-07-07	50000
203	Koko	2020-04-15	70000
204	Adarsh	2025-09-25	80000
205	Shreya	2026-06-22	40000

14. Rename a table.

Solution:

```
75 •  alter table Employee rename emp1;
76 •  select*from emp1;
```

Result Grid | Filter Rows: | Edit:

Emp_id	Emp_name	City	bdate
102	Khush	Dubai	2020-09-11
103	Koko	Chandrapur	2006-06-26
104	Adarsh	Nagpur	2011-07-17
105	Shreya	Indore	2004-02-16
NULL	NULL	NULL	NULL

15. Delete all the records using the TRUNCATE command.

Solution:

truncate table emp1;

```
158  16:01:48  truncate table emp1   0 row(s) affected   0.031 sec
```

16. Drop both tables from the database.

Solution:

drop table emp1,Investor;

```
159  16:03:21  drop table emp1,Investor   0 row(s) affected   0.015 sec
```